

# Solving Equations

WorksheetHub AI  
Mathematics · Grade 7

Unit 2 · Algebra

Medium

Solving linear equations

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Score: / 16

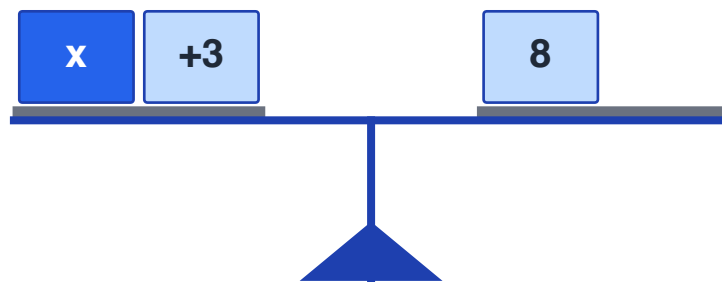


## Remember

To solve an equation, **undo** the operations using the **opposite** (inverse) operation, in reverse order, keeping the equation balanced.

+ and – are opposites.  $\times$  and  $\div$  are opposites.

$x + 3 = 8$  then take 3 from BOTH sides then  $x = 5$



An equation is a balance. Whatever you do to one side, do to the other.



## Worked example

Solve  $3x + 2 = 17$

Step 1: subtract 2 from both sides  $\rightarrow 3x = 15$

Step 2: divide both sides by 3  $\rightarrow x = 5$

Check:  $3 \times 5 + 2 = 17$  ✓

## A. One-step equations

1  $x + 7 = 15$

Answer: \_\_\_\_\_

2  $y - 5 = 9$

Answer: \_\_\_\_\_

3  $4a = 28$

Answer: \_\_\_\_\_

4  $\frac{b}{3} = 6$

Answer: \_\_\_\_\_

## B. Two-step equations

5  $3x + 2 = 17$

Answer: \_\_\_\_\_

6  $5y - 4 = 21$

Answer: \_\_\_\_\_

7  $\frac{x}{2} + 3 = 9$

Answer: .....

8  $7 + 2m = 19$

Answer: .....

### C. Equations with brackets

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9  $2(x + 3) = 18$

Answer: .....

10  $3(2x - 1) = 21$

Answer: .....

11  $4(y - 2) = 20$

Answer: .....

### D. Word problems

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12 I think of a number, double it and add 5. The answer is 17. What is my number?

Answer: .....

13 A taxi charges \$4 plus \$3 per km. A trip costs \$19. How many km was the trip?

Answer: .....

14 The perimeter of a rectangle is 32 cm. Its length is  $x + 4$  and its width is  $x$ . Find  $x$ .

Answer: .....

### ★ Challenge

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15 Solve  $5x - 3 = 2x + 9$  ( $x$  is on both sides).

Answer: .....

16 Solve  $2(3x - 4) = 4x + 6$ .

Answer: .....